Gender Imbalance, Marriage Stability, and Divorce Rate: Evidence from China*

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Abstract

The deficit of men or women in a regional marriage market is commonly observed. However, literature on the impact of this deficit on marriage stability reports mixed results, with little evidence on the underlying mechanisms. In this paper, using provincial, census, and household survey data in China, we find that a higher male-to-female ratio increases divorce rates. Further analyses suggest that this impact is primarily driven by married women having more options outside their marriage. These findings highlight the significance of gender balance in sustaining stable marriages and uncover a new contributing factor to the escalating divorce rates in China.

Keywords: Divorce Rate, Gender Imbalance, Outside Option
JEL Codes: D10, J12, J16, N35

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1 Introduction

China's crude divorce rate, defined as the number of annual divorces per 1,000 people, surpassed that of the United States in 2017 and has remained higher since, drawing significant government attention. This surge has coincided with the marriage and divorce of the birth-control generation who face a highly imbalanced sex ratio, with significantly more men than women in the marriage market. This pattern is not unique to China; rising divorce rates have also coincided with gender imbalances in other developing countries, such as Vietnam and Thailand (OECD and Centre, 2021). Given the importance of sex ratios in shaping marriage market outcomes (e.g., Becker, 1981), this paper examines whether the skewed sex ratio has contributed to China's escalating divorce rate.

The theoretical relationship between the sex ratios and divorce is ambiguous. On the one hand, the previous literature shows that a shortage of women in the marriage market can benefit them by increasing their chances of marrying up, raising bride prices, and enhancing their bargaining power within households (see, e.g., Francis, 2011; **?**). This would theoretically improve women's marital satisfaction and thus increase marital stability. On the other hand, the gender imbalance can expand married women's outside options by increasing the pool of potential partners, which may prompt them to leave their current marriages. Appendix Section A presents a simple model showing that, under different conditions, the same framework can yield either positive or negative effects of sex imbalance on divorce rates.

There are multiple reasons why more "outside options" may increase divorce. First, couples enter marriage with incomplete information (Becker et al., 1977), and a low realized utility, such as an abusive husband, may prompt women to seek outside options. Second, changes in the choice set, like new workplace contacts, can lead to divorce if a woman meets someone she views as a better match (McKinnish, 2004). Additionally, individuals' preferences for partners may change over time. In all the above scenarios, a greater number of available men increases the options for women seeking to improve their marital situation. The outside option channel has rarely been empirically examined, possibly due to data limitations. Our data uniquely provides information on remarriage and the quality of the subsequent husbands, allowing us to address this gap.

China provides a compelling setting to study the relationship between the sex ratios and divorce due to its large population and persistent gender imbalance. The country has one of the highest sex ratios at birth in the world, with 121.7 boys per 100 girls in rural areas in 2000 (Li, 2007). This imbalance stems largely from the son preference and the One-Child Policy (OCP), which led to sex selection after its introduction in the late 1970s. Although it officially ended in 2016, the policy has affected the births over the past 30 years and will continue to shape the marriage market for decades.

Despite being commonly known as the "one-child policy," additional births were not always prohibited. In practice, enforcement varied considerably depending on local socioeconomic conditions, governance capacity, and administrative discretion (Qian, 2009; Ebenstein, 2010; Liu, 2014). As a result, the degree of gender imbalance differed widely across both geography and birth cohorts. This extensive variation generates quasi-experimental conditions that we exploit to identify the causal effect of the sex ratios on divorce.

We begin by establishing that a surplus of males in the marriage market increases the divorce rate. At the aggregate level, provinces with higher male-to-female ratios exhibit significantly higher annual divorce rates. At the individual level, using the census sample data, we construct the sex ratios at the prefecture-by-birth-year level and examine their impact on divorce, controlling for individual characteristics as well as birth year and prefecture fixed effects. We find that a higher sex ratio significantly increases the likelihood of women being currently divorced.

We test the robustness of our findings using a two-stage least squares (2SLS) approach with multiple instrumental variables. We also rule out alternative explanations such as changes in age at marriage, cohabitation with in-laws, and the gender education gap. Moreover, no major changes in divorce law occurred during our sample period that would confound the results.[1]

The census data records only current marital status and misses divorce history for remarried women. To address this, we use marriage history data from the 2010 China Family Panel Studies (CFPS) to construct an "ever divorced" indicator. We find that higher sex ratios increase the likelihood of women having experienced divorce.

We then show, consistent with the existing literature, that a higher sex ratio improves women's marriage quality. This suggests that the impact of the sex ratio on divorce is not driven by women's difficulty in finding suitable first-marriage partners. Women exposed to higher sex ratios tend to marry men with better education, higher income, and urban *Hukou*, which is linked to better social security. They also live in higher-quality housing, with larger areas, more rooms, and higher prices. In addition, they report lower levels of depression within marriage.

Further evidence supports the importance of the outside option channel. We find that divorced or widowed women in markets with higher sex ratios are more likely to

[1]During our sample period (2005–2015), the only notable legal change was the 2011 "Judicial Interpretation III," which weakened women's post-divorce property rights. Since our regressions control for calendar year and birth year fixed effects, the nationwide policy implementation should not affect the results.

remarry, and those who do tend to have higher-quality partners. A higher likelihood of remarriage and the prospect of a relatively better subsequent husband can reduce the reluctance to divorce, helping explain the higher divorce rates in the male-skewed markets.

Additionally, the results suggest that the higher male-to-female ratio reduces men's marital surplus. Men exposed to higher sex ratios face higher divorce rates, spend more time caring for family members on weekdays, likely due to the increased pressure to contribute at home, and report greater psychological distress.

This paper contributes to the literature on the determinants of divorce. The existing research has examined factors such as education (e.g., Bruze et al., 2015), age at marriage (Teachman, 2002; Rotz, 2016), income (e.g., Bertrand et al., 2015), and the presence or gender of children (e.g., Dahl and Moretti, 2008; Mammen, 2008). However, while many developing countries now experience both high sex ratios and rising divorce rates, there is limited empirical evidence on the effect of gender imbalance, particularly a shortage of women, on divorce.

The few existing studies on gender imbalance and divorce primarily focus on developed countries and examine the symmetric case of a shortage of men, often resulting from high male mortality during wars. These studies report mixed results. Abramitzky et al. (2011) find that the post–World War I male shortage in France reduced divorce rates, while Brainerd (2017) and Ogasawara and Igarashi (2021) show that male scarcity after World War II increased divorce rates in Russia and Japan, respectively. Identifying the causal effect using war is difficult, as war also affects mental health and labor markets, both of which influence marriage outcomes (Negrusa et al., 2014). Beyond war contexts, Lafortune (2013) finds that immigration-driven increases in the male-to-female ratio destabilize marriages for men, while the effect for women is insignificant, likely due to measurement limitations and limited statistical power. This paper contributes by studying a different setting, leveraging a distinct source of variation in the sex ratio, applying a better measure of divorce, and examining the underlying mechanisms.

In addition, this paper contributes to the broad literature on gender imbalance (Becker, 1981), especially the consequence of "Missing Women" (Sen, 1992). The existing studies show that a shortage of women increases their likelihood of marrying up (Abramitzky et al., 2011; Du et al., 2015; Battistin et al., 2022), raises bride prices (Nick and Walsh, 2007; Francis, 2011; Grossbard, 2015), drives grooms or their parents to save more and invest in high-value housing (Wei and Zhang, 2011a; Wei et al., 2017; Nie, 2020; Tan et al., 2021), and encourages greater pre-marital investments by men (Lafortune, 2013). As a natural extension, sex imbalance may also influence marriage stability. This paper shows that the shortage of women partly drives the escalating divorce rates in China.

2 Data and Descriptive Patterns

2.1 Provincial Panel Data

The province-level analyses use data from the China Statistical Yearbooks (1997–2020), which provide annual information on the number of divorce cases, demographic composition, and socioeconomic characteristics such as GDP per capita. To measure the sex ratio in the marriage market, we use the 2000 census sample, inferring provincial sex ratios based on birth cohorts. For example, we use the sex ratio of those aged 15–35 in 2000 as a proxy for the 20–40 population in 2005.

In the provincial analyses, the main outcome is the number of divorces per 1,000 people aged 15 to 64. We use alternative measures such as the divorce-to-marriage ratio for robustness tests. The sex ratio in the marriage market is defined as the male-to-female ratio among the population aged 20–40. More than 90% of women who divorced got divorced when they were of this age range, according to the China Family Panel Studies (CFPS) data. The descriptive statistics are reported in Appendix Table B.1, Panel A.

2.2 Census Data

For the individual-level analyses, we use the census sample data from 2005, 2010, and 2015, which include each woman's current marital status. In 2010 and 2015, marital status is classified as never married, married, divorced, or widowed. The 2005 data further distinguishes between first marriages and remarriages. The datasets also report prefecture of residence and various demographic and socioeconomic characteristics, including fertility, education, and employment. We focus our analyses on the currently married or divorced women aged 20-40.

We measure the sex ratio following the previous literature (Brainerd, 2017; Ogasawara and Igarashi, 2021). For women living in prefecture c born in year y, the sex ratio is constructed in the following way:

$$Sex\_ratio_{cy} = \frac{\sum_{t=y-8}^{y+2} \# \, of \ men \ born \ in \ year \ t \ in \ prefecture \ c}{\sum_{t=y-8}^{y+2} \# \, of \ women \ born \ in \ year \ t \ in \ prefecture \ c} \tag{1}$$

To address potential endogeneity from gender-biased local labor market shocks, we construct the sex ratio measure using the 2000 census, prior to the sample period. We use a birth year gap of -8 to +2, reflecting that women typically marry men older than themselves. To reduce noise from small prefecture-year cells, we winsorize the sex ratio at the 1st and 99th percentiles. We consider prefectures to be the relevant geographic unit for the marriage market based on the previous literature (Wei et al., 2017). The sex ratio is then merged with census data from 2005, 2010, and 2015, based on each woman's prefecture of residence and birth year.

The key variables are summarized in Appendix Table B.1 Panel B. The natural sex ratio at birth is typically around 1.05–1.06 (Sen, 1992), but the ratio among marriage-age populations is usually lower due to factors such as mortality, war, and aging (Ritchie and Roser, 2019). In our census sample, the overall sex ratio is 1.03. However, this masks variation by birth cohort: for those born before the One-Child Policy, the average is 1.01, while for those born after, it rises to 1.08. The disparity highlights the impact of the birth control policy on the sex ratio.

2.3 CFPS 2010

We use the 2010 wave of the China Family Panel Studies (CFPS) to complement the census data. The CFPS is a nationally representative longitudinal survey conducted by the Institute of Social Science Survey at Peking University. It covers 25 provinces, representing approximately 94.5% of China's population.[2]

The survey data has two key advantages. First, it provides detailed marriage histories, including the year of each marriage and divorce and the total number of marriages, enabling us to construct an "ever divorced" indicator rather than relying solely on current status. Second, it includes rich information on household and individual characteristics such as number of siblings, income, and psychological well-being. The rich information allows us to account for additional individual characteristics, further addressing potential biases related to omitted variables.

For the sex ratio, we use the same data as in the main census analyses. The sex ratios are merged with the survey data based on each woman's prefecture of residence and birth year. Key variables are summarized in Appendix Table B.1, Panel C.

[2]The excluded provinces/cities/autonomous regions in mainland China are Xinjiang, Tibet, Qinghai, Inner Mongolia, Ningxia, and Hainan.

3 Main results

3.1 Gender Imbalance and Divorce Rate: Provincial Panel Data

For each province p in year t, we estimate the following equation:

$$Divorce\_rate_{pt} = \beta_0 + \beta_1 Sexratio20\_40_{pt} + X_{pt}\eta + \gamma_p + \mu_t + \epsilon_{pt} \tag{2}$$

where $Sexratio20\_40_{pt}$ is the male-to-female ratio of population aged 20-40 in year t in province p. X_{pt} is a vector of control variables, including GDP per capita, average household size, and average years of education. γ_p and μ_t are province fixed effects and year fixed effects. With only 31 provinces in mainland China, there is a potential concern that there may not be enough clusters for the asymptotics (Donald and Lang, 2007). To address this issue, we conduct the wild cluster bootstrap test, as suggested by Cameron et al. (2008), and report the p-values in square brackets.

Table 1 shows the results. Across all regressions, the sex ratio coefficients are positive and significant, indicating a strong positive relationship with the divorce rate. The magnitudes of the coefficients are non-negligible. For example, the coefficient in Column (4) implies that a one-standard-deviation increase in the sex ratio raises the divorce rate by 11.3 percent. The results remain robust when using the divorce-to-marriage ratio as the outcome variable. Calculated as the annual divorces divided by marriages and then multiplied by 100, this measure accounts for variations in marriage rates.

3.2 Gender Imbalance and Women's Probability of Divorce: Census Data

We now turn to the individual-level analyses, which allow for richer controls and define marriage markets at the prefecture level. For each woman i in prefecture c, born in year y, and surveyed in wave w, we estimate the following equation:

$$Divorce_{icyw} = \alpha_0 + \alpha_1 Sex\_ratio_{cy} + Z_{icyw}\eta + \delta_{cw} + \kappa_{yw} + \epsilon_{icyw} \tag{3}$$

where Z_{icyw} is a vector of controls including indicators for having a son, Han ethnicity, employment status, migrant status, urban residence, and fixed effects for education level and number of children. δ_{cw} denote the prefecture times census wave fixed effects and κ_{yw} denote the birth year times census wave fixed effects. Standard errors are clustered at the prefecture level.

Table 1: Gender Imbalance and Divorce Rate at Provincial Level

	Divorce Cases Divided by Population Aged 15-64				Divorce Cases Divided by Marriage Cases
	(1)	(2)	(3)	(4)	(5)
Sexratio20_40	8.114**	6.212**	5.504*	5.516*	57.099*
	(2.953)	(3.013)	(2.906)	(3.018)	(28.999)
	[0.012]	[0.060]	[0.088]	[0.097]	[0.065]
ln(GDP per capita)		0.838	0.891	0.885	-11.906**
		(0.556)	(0.553)	(0.563)	(5.164)
Household size			0.340*	0.342*	9.582***
			(0.172)	(0.194)	(1.945)
Years of education				0.015	-0.986
				(0.288)	(1.642)
Obs.	744	744	744	744	651
Adj. R-sq	0.878	0.880	0.880	0.880	0.869
Mean of Dep. Var.	2.503	2.503	2.503	2.503	28.449

NOTE: Standard errors clustered at province level are reported in parentheses. The entries in square brackets are $p - values$ from the wild cluster bootstrap test with the null hypothesis of the coefficient equal to zero. ***p<0.01, **p<0.05, *p<0.1.

There are substantial variations in the sex ratios across prefectures and birth cohorts, reflecting differences in policy enforcement. Figure 1 plots the residualized sex ratios, obtained by regressing the sex ratios on prefecture and birth year fixed effects, for six randomly selected prefectures. Two key patterns emerge. First, there is no consistent pattern across prefectures. The sex ratios rose and fell at different times, with no persistent trend in most prefectures. Second, most prefectures exhibit considerable variation over time. A change of 0.05 corresponds to five additional men per 100 women.

The main results are reported in Table 2. The sex ratio coefficient is positive and statistically significant when all census waves are pooled. A coefficient of 0.020 implies that a one-standard-deviation increase in the sex ratio is associated with a 9.6 percent higher probability of being currently divorced. When analyzing each census wave separately, the sex ratio coefficient is positive and marginally significant in 2005, but becomes larger and more significant in the 2010 and 2015 waves. This likely reflects the relative rarity of divorce in the early 2000s. As divorce became more common over time, variation across prefectures and cohorts increased, making the effect of the sex ratio more detectable in later years.

We conduct a series of robustness checks, with results reported in Appendix Table

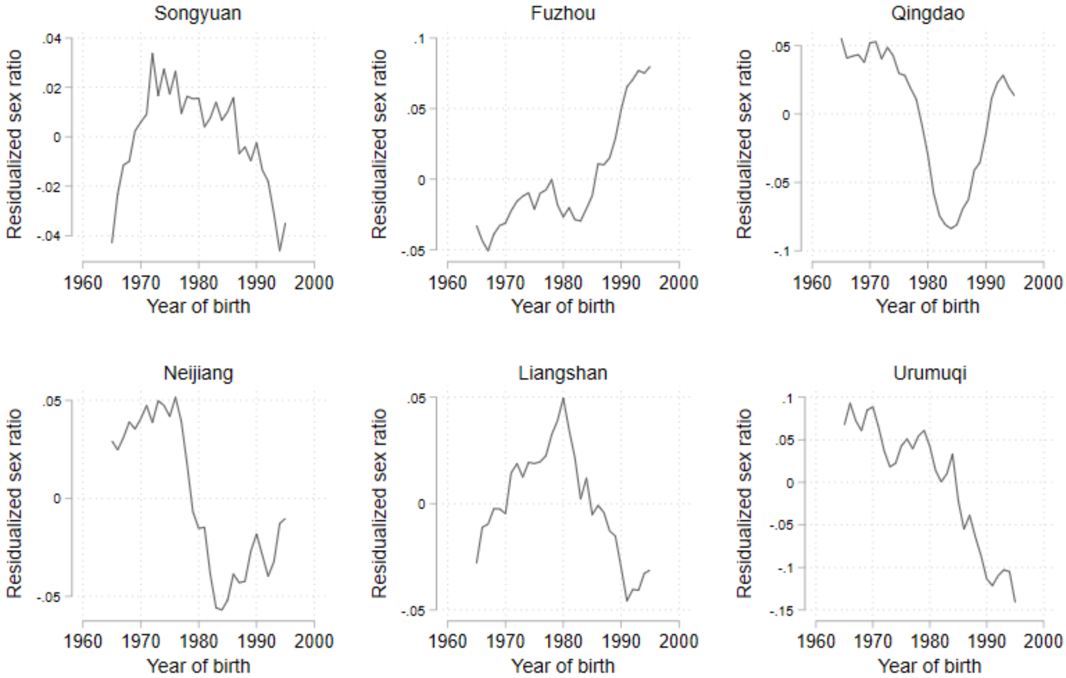

Figure 1: Residualized Sex Ratios

NOTE: The figure shows the residualized sex ratios obtained from regressing the sex ratios on prefecture fixed effects and birth year fixed effects.

B.2. First, we alter the age range to 25–45 to test sensitivity to age selection. Second, we adjust the age cohorts used to calculate the sex ratio, using alternative ranges such as [–6,+1] and [–8,+2]/[–5,+5].[3] Third, we replace the baseline 2000 sex ratios with wave-specific sex ratios constructed from each census wave. Across all specifications, the results consistently support a positive relationship between the sex ratios and the probability of being currently divorced.

We take several measures to address potential identification concerns, with results reported in Appendix Table B.3. First, higher sex ratios could affect women's independence through education and labor market outcomes, potentially influencing divorce. While we control for women's education and employment in the main analysis, the results remain robust when further controlling for industry and occupation fixed effects, suggesting that the increased independence is unlikely to drive the findings. Second, we show that changes in social attitudes toward divorce are unlikely to explain the results,

[3] [–6,+1] refers to $Sex\_ratio_{cy} = \frac{\sum_{t=y-6}^{y+1} of\ men\ born\ in\ year\ t\ in\ prefecture\ c}{\sum_{t=y-6}^{y+1} of\ women\ born\ in\ year\ t\ in\ prefecture\ c}$, and [–8,+2]/[–5,+5] refers to $Sex\_ratio_{cy} = \frac{\sum_{t=y-8}^{y+2} of\ men\ born\ in\ year\ t\ in\ prefecture\ c}{\sum_{t=y-5}^{y+5} of\ women\ born\ in\ year\ t\ in\ prefecture\ c}$

Table 2: Gender Imbalance and Women's Probability of Divorce: Censuses

	Pooled (1)	Census 2005 (2)	Census 2010 (3)	Census 2015 (4)	IV (5)
Sex Ratio	0.020***	0.010*	0.023***	0.020***	0.099**
	(0.004)	(0.006)	(0.005)	(0.007)	(0.041)
Han ethnicity	-0.003***	-0.005***	-0.003**	-0.001	-0.002
	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
Having a son	-0.005***	-0.004***	-0.006***	-0.006***	-0.005***
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)
Employed	0.001**	-0.002**	0.001**	0.003***	-0.000
	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)
Migrant	-0.003***	-0.005***	-0.002***	-0.003***	-0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Urban	0.008***	0.009***	0.009***	0.005***	0.009***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
1st-Stage F-stat					6.393
Observation	1,150,344	339,405	593,688	217,250	461,665
Adj. R-sq	0.018	0.017	0.018	0.016	0.000
Mean of Dep. Var.	0.016	0.011	0.018	0.020	0.018

NOTE: All columns control for prefecture-by-wave, birth year-by-wave, education, and number of children fixed effects. Standard errors clustered at the prefecture level are shown in parentheses. ***p<0.01, *p<0.05, p<0.1.

as the results remain robust when controlling for province-by-birth-year fixed effects. Third, as noted by Chen et al. (2021) and Alm et al. (2022), some couples engage in "fake divorces" to obtain additional housing purchase quotas. Since the sex ratios may influence housing demand, this could confound the results. However, controlling for housing quota policies does not change the estimates. Finally, because we observe the sex ratios based on current residence, not necessarily where divorce decisions were made, we exclude migrants to address this concern. The results remain robust.

Given the long time span of over twenty years, it is unlikely that parents engaged in sex selection based on anticipated future marriage market conditions. The main concern for identification is thus omitted variable bias. In Section 3.4, we examine and rule out the major confounding factors. Importantly, our sex ratio measure is highly granular, and we control for both prefecture and birth year fixed effects. As a result, any remaining threats to identification would need to vary at the prefecture-by-birth-year level.

Finally, we experiment with multiple instruments to address concerns about potential omitted unobservables. However, the high granularity of the sex ratio measure leads to weak first stages, making 2SLS estimates less precise and 1–5 times larger than the OLS estimates. To avoid issues related to weak instruments, our main analyses rely on ordinary least squares (OLS). The preferred 2SLS result, reported in Table 2, Column (5), uses an interaction between prefecture-level gender norms constructed using the

2010 Chinese General Social Survey (CGSS) and an indicator for being born in or after 1979, the start of the One-Child Policy. The sex ratio coefficient remains positive and significant. Appendix Section B.3 provides details on alternative instruments and further discussion.

3.3 Gender Imbalance and Women's Probability of Divorce: Individual Survey Data

The census records only current marital status, so women who divorced and remarried are coded as "married," leading to an underestimation of the sex ratio's impact on divorce. To address this, we use the CFPS data, which includes complete marriage histories, and estimate the following equation:

$$Ever\_divorced_{icy} = \alpha_0 + \alpha_1 Sex\_ratio_{cy} + Z_{icy}\eta + \delta_c + \kappa_y + \epsilon_{icy} \tag{4}$$

where Z_{icy} is a vector of individual characteristics. In addition to the controls used in the regressions using census data, we include an indicator for pre-marital cohabitation and fixed effects for the number of male and female siblings. The standard errors are clustered at the prefecture level.

Table 3 reports the regression results. Even with extensive controls and fixed effects, the sex ratio coefficients remain positive and significant, indicating that higher sex ratios increase the likelihood of divorce. A coefficient of 0.098 implies that a one-standard-deviation increase in the sex ratio raises the probability of ever divorcing by 23 percent. The larger effect size, compared to the estimates using the census data, is consistent with the census underestimating divorce because it records remarried women as currently married.

Additionally, we show that sibling structure does not confound the relationship between the sex ratio and the divorce rate. The birth control policy may simultaneously affect the sex ratio, sibling composition, and number of siblings. The previous research on the "little prince/princess" phenomenon suggests that only children may be spoiled, potentially shaping their expectations of marriage and influencing their divorce decisions (Cameron et al., 2013). However, as shown in Column (3), the positive effect of the sex ratio on divorce remains robust, and the coefficient changes little after controlling for fixed effects for the number of male and female siblings.

Table 3: Gender Imbalance and Women's Probability of Divorce: CFPS 2010

1{Ever divorced}	(1)	(2)	(3)	(4)
Sex Ratio	0.085*	0.098**	0.098**	0.098**
	(0.047)	(0.046)	(0.047)	(0.046)
Controls	-	Y	Y	Y
Male and female siblings FE	-	-	Y	Y
Income	-	-	-	Y
Obs.	8,799	8,798	8,717	8,715
Adj. R-sq	0.019	0.073	0.074	0.075
Mean of Dep. Var.	0.028	0.028	0.028	0.028

NOTE: All columns control for fixed effects for prefecture, birth year, education, and number of children. Columns (2)–(4) include all controls from the census regressions, along with an indicator for pre-marital cohabitation. Column (4) additionally controls for the inverse hyperbolic sine of income. Using log income instead slightly increases the coefficient without affecting significance, though the sample size drops to about three-quarters. Standard errors clustered at the prefecture level are reported in parentheses. ***p<0.01, *p<0.05, p<0.1.

3.4 Rule Out Other Potential Explanations

First, a higher sex ratio may lead women to marry earlier by increasing their exposure to potential partners at a younger age. Early marriage is associated with a higher risk of divorce due to lower maturity and conflict resolution skills (Rotz, 2016). To test this, we add fixed effects for age at first marriage to the baseline specification. As shown in Table 4, Column (2), the sex ratio coefficient remains positive, significant, and similar in magnitude to the baseline result in Column (1). This suggests that age at first marriage does not explain the effect. In fact, results indicate that the higher sex ratios are associated with a slight increase in marriage age.

Second, birth control policies not only affected the sex ratios but also increased the prevalence of one-child households. In China, especially in rural areas, husbands who are only children are more likely to co-reside with their parents, which raises the likelihood that their wives live with in-laws. This can increase divorce rates by intensifying conflict, caregiving burdens, and loss of privacy. To address this, we calculate the co-residence rate at the prefecture-by-birth-year level for each census wave, separately for urban and rural areas, and include it as a control. Results are reported in Table 4, Column (3). The number of observations decreases due to the exclusion of prefecture–birth year–urban cells with limited data. The sex ratio coefficient remains positive and significant, suggesting that co-residence is not a major source of bias, though it does significantly increase women's likelihood of divorce.

Third, birth control policies may raise educational attainment for both men and women, with potentially larger gains for women, thereby narrowing the gender education gap (Chae et al., 2023). Since the impact of a narrower education gap on marriage stability is debated in the literature (Bertrand et al., 2015; Binder and Lam, 2022), we control for it in our analysis. For each woman observed in census wave w, living in prefecture c and born in year y, we compute the average male–female education gap by census wave, prefecture, and birth year. We then average the gaps over birth years $[y-8, y+2]$ to approximate the gap she faced at marriage. Results are shown in Table 4, Column (4). The sex ratio coefficient remains positive and significant, indicating that the dynamics of the education gap do not drive the main findings.

Finally, when we include all controls in the same regression, the coefficient for the sex ratio is positive and significant and has a similar magnitude as the main results.

Table 4: Rule Out Other Explanations: Censuses

	Baseline	Marriage age	Fraction of wives co-living with husband's parent	Gender Edu Gap	Add all controls
	(1)	(2)	(3)	(4)	(5)
Sex Ratio	0.020***	0.021***	0.021***	0.020***	0.022***
	(0.004)	(0.005)	(0.008)	(0.004)	(0.008)
Co-residence			0.022***		0.025***
			(0.003)		(0.003)
Education gap				-0.004***	-0.004**
				(0.001)	(0.002)
Marriage age FE	-	Y	-	-	Y
Obs.	1,150,344	933,093	795,372	1,150,344	695,204
Adj. R-sq	0.018	0.019	0.019	0.018	0.020
Mean of Dep. Var.	0.016	0.016	0.017	0.016	0.016

NOTE: The co-residence rate is calculated at the prefecture-by-birth-year level for each census wave, separately for urban and rural areas. The gender education gap is computed as follows: for each woman observed in wave w, living in prefecture c and born in year y, we first calculate the average male–female education gap by census wave, prefecture, and birth year. We then average the gaps over birth years $[y-8, y+2]$ as a proxy for the education gap she faced at marriage. All columns control for prefecture-by-wave, birth year-by-wave, education, and number of children fixed effects, as well as the full set of controls from the main census regressions. Column (1) replicates the baseline results from Table 2. Standard errors clustered at the prefecture level are in parentheses. ***p<0.01, *p<0.05, p<0.1.

Another possible explanation is that the gender imbalances increase mating competition, prompting men to work longer hours or migrate for jobs, as noted by Wei and Zhang (2011b) and Zhang et al. (2021). However, if such efforts reduce their presence at home or strain relationships, this could increase divorce rates. We address this concern in two ways. First, the next section shows that women report higher marital satisfaction, which contradicts the idea that they are unhappier due to their husbands' absence or relationship shortcomings. Second, we calculate the share of male migrant workers by census year, prefecture, and birth year. Controlling for this measure does not alter our results.

4 Mechanism

Overall, we find a robust pattern that the higher sex ratios increase women's probability of divorce. To better understand this relationship, we explore the potential underlying channels.

The effect of the sex ratios on divorce is theoretically ambiguous. A higher sex ratio may improve women's bargaining power, allowing them to marry better partners and reducing their risk of divorce. Conversely, it may expand outside options, making marriage less stable. Appendix Section A presents a simple model showing that both effects can arise within the same framework, depending on the underlying conditions. In the following sections, we empirically examine both mechanisms.

4.1 Do Women Marry Up When the Sex Ratio is Higher?

If excessive frictions prevent women from finding better partners, or if the prospect of remarriage lowers selectiveness in first marriages, a higher sex ratio may not improve match quality. To test this, we regress measures of husband quality on the sex ratio.

The results are reported in Table 5 Column (1)-(3). To limit survivorship bias, we restrict the sample to couples that have been married for three years or less. The findings indicate that women exposed to higher sex ratios are more likely to marry men with more education and higher income. Women with rural *Hukou* are more likely to marry men with urban *Hukou*, which provides better social security benefits.

Given the importance of housing in marriage, we also examine how the sex ratios affect housing quality. In China, where parents-in-law often contribute to marriage-related house purchases, housing characteristics can proxy for parental affluence. Columns (4)–(6) of Table 5 show that the higher sex ratios are associated with larger housing area, more rooms, and higher house prices.

Table 5: Gender Imbalance and Husband Quality

	Husband Characteristics			Housing Quality		
	Education	log Monthly Income	Has Urban Hukou	lnArea	# of Rooms	lnHouseprice
	(1)	(2)	(3)	(4)	(5)	(6)
Sex Ratio	0.585**	0.277**	0.105***	0.779***	1.175***	0.298*
	(0.236)	(0.113)	(0.039)	(0.174)	(0.275)	(0.156)
Obs.	103,264	28,549	72,238	103,188	103,188	24,226
Adj. R-sq	0.627	0.536	0.152	0.312	0.256	0.575
Mean of Dep. Var.	10.684	6.483	0.083	4.376	3.200	10.227

NOTE: To limit survivorship bias, the sample includes only couples married within three years. Since the 2015 Census lacks information on marriage years, only the 2005 and 2010 waves are used. Columns (2) and (6) use Census 2005 only due to data availability, and Column (3) is restricted to women with rural *Hukou*. All columns control for prefecture-by-wave, birth year-by-wave, and wife's education fixed effects. Columns (4)–(6) also control for fixed effects for the year the house was built. All census control variables are included in all columns, except for the number of children and the indicator for having a male child, which are excluded because fertility occurs post-marriage and is unlikely to affect husband or housing characteristics directly. Including these variables yields similar results. Standard errors clustered at the prefecture level are reported in parentheses. ***p<0.01, *p<0.05, p<0.1.

Table 6: Gender Imbalance and Wife's Mental Health

The frequency of feeling	Upset	Stressed	Cannot calm down	Hopeless	Everything is difficult	Life is meaningless	Overall depression
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Sex Ratio	-0.624***	-0.117	-0.425**	-0.256*	-0.448**	-0.207	-0.957**
	(0.236)	(0.197)	(0.186)	(0.152)	(0.203)	(0.173)	(0.449)
Obs.	8,558	8,565	8,564	8,545	8,565	8,558	8,530
Adj. R-sq	0.056	0.046	0.056	0.051	0.082	0.060	0.088
Mean of Dep. Var.	0.763	0.569	0.493	0.325	0.562	0.333	-0.014

NOTE: This table presents estimates using CFPS data, restricted to currently married women. Outcomes in Columns (1)–(6) are indices measuring the frequency of depressive symptoms, ranging from 0 (none of the time) to 4 (all of the time). Column (7) reports the first principal component from a Principal Component Analysis (PCA) of these measures. All columns control for fixed effects for prefecture, birth year, education, and number of children, as well as all census regression controls and an indicator for pre-marital cohabitation. Standard errors clustered at the prefecture level are reported in parentheses. ***p<0.01, *p<0.05, p<0.1.

Moreover, we examine whether women report fewer mental health issues in marriage when the sex ratio is higher, addressing concerns that unobserved traits like personality also influence marital quality. We use women's self-reported psychological distress from the CFPS data, measured by the Kessler 6 Scale (K6) (Kessler et al., 2002). The survey asked about the frequency of feeling "upset", "stressed", "cannot calm down", "hopeless", "everything is difficult", and "life is meaningless," with responses coded from 0 (none of the time) to 4 (all of the time). We regress the distress indexes on the sex ratio for currently married women. As shown in Table 6, all coefficients are negative, indicating that higher sex ratios are associated with lower levels of psychological distress. The results remain robust after controlling for women's income.

In conclusion, women exposed to higher sex ratios tend to marry better, suggesting that higher divorce rates cannot be explained solely by poor initial matches.

4.2 Outside Option

4.2.1 Do Women Remarry More Easily after Divorce?

We first examine whether divorced women are more likely to remarry when faced with higher sex ratios. If divorce reflects a declining demand for marriage, the sex ratios should not affect remarriage. In other words, if we do find an effect, it suggests that the availability of alternative options may play a role in women's divorce decisions. The anticipation of easy remarriage could reduce the reluctance to divorce, contributing to higher divorce rates in more male-skewed markets.

Table 7 presents the estimated effects of the sex ratio on remarriage using the CFPS data. We examine both ever-divorced and ever-widowed women, as remarriage can follow either of these events. In both samples, the higher sex ratios are associated with a greater likelihood of remarriage. These findings suggest that women face better prospects for finding a new partner in male-skewed markets, which may increase their willingness to divorce.

4.2.2 Is the Quality of the Subsequent Husband Better for Remarried Women?

We then examine whether the higher sex ratios improve the quality of subsequent partners for remarried women. If so, it suggests that the availability of attractive alternatives influences divorce decisions. Women anticipating a relatively higher-quality future partner may be more willing to leave an unsatisfying marriage.

Table 8 shows the results. The sample includes all currently married or remarried women. The coefficient on the remarriage indicator shows that remarried women tend to

Table 7: Gender Imbalance and Remarriage

1{Ever-remarry}	Divorced	Divorced & Widowed
	(1)	(2)
Sex Ratio	3.267**	2.550***
	(1.393)	(0.933)
Obs.	201	400
Adj. R-sq	0.258	0.139
Mean of Dep. Var.	0.507	0.450

NOTE: All columns control for fixed effects for prefecture, birth year, education, number of children, and number of male and female siblings, as well as all census regression controls and an indicator for pre-marital cohabitation. Standard errors clustered at the prefecture level are reported in parentheses. ***p<0.01, *p<0.05, p<0.1.

have partners with fewer years of education and lower income. However, rural women are more likely to remarry men with urban *Hukou*. Overall, the results suggest that remarried women typically marry lower-quality partners than those in first marriages. There are at least two potential explanations. Firstly, divorced women may be less preferred in the marriage market, attracting lower-quality matches. Secondly, higher-quality men may be more likely to remain in their first marriage. As a result, partner quality may improve relative to a prior bad match but remain lower on average than that of first-married women.

Our results suggest that, conditional on remarriage, a higher sex ratio mitigates the decline in husband quality. Women in male-skewed markets remarry men with higher education, and rural women are more likely to remarry men with urban *Hukou*.

The magnitudes of the coefficients on the remarriage indicator and its interaction with the sex ratio suggest that, for education and income, higher sex ratios do not fully offset the costs of remarriage. Even in markets with the highest sex ratios, remarriages remain less favorable than first marriages. This implies that, for many women, divorce is likely a means of escaping deeply unsatisfactory or abusive relationships rather than a strategy for economic advancement.

5 How about Men?

The preceding analyses and the model in Appendix A focus on women, treating men as passive participants in marriage and divorce decisions. This section shifts the focus to men, showing that higher sex ratios reduce their marital surplus.

Table 8: Gender Imbalance and Quality of the Subsequent Husband

Husband's	Year of Education	log Monthly Income	Has Urban Hukou
	(1)	(2)	(3)
Sex Ratio_demeaned	-0.120	0.103	0.026
	(0.147)	(0.068)	(0.020)
1{Remarry}	-0.337***	-0.050***	0.042***
	(0.034)	(0.012)	(0.009)
Sex Ratio_demeaned	1.057**	0.134	0.305**
×1{remarry}	(0.531)	(0.192)	(0.146)
Obs.	207,826	197,690	149,283
Adj. R-sq	0.534	0.471	0.081
Mean of Dep. Var.	9.452	6.370	0.053

NOTE: This table uses the Census 2005 data. Column (3) is restricted to women with rural *Hukou*. All columns control for fixed effects for prefecture, birth year, and education. Census regression controls are included in all columns, except for the number of children and the indicator for having a male child. Standard errors clustered at the prefecture level are reported in parentheses. ***p<0.01, *p<0.05, p<0.1.

We begin by examining the impact of the sex ratio on men's divorce rates. As same-sex marriage is not recognized in China, each divorce involves both a man and a woman. Given our finding that higher sex ratios raise divorce rates among women, we expect a similar pattern for men. We replicate the analysis for men aged 20–50 (compared to 20–40 for women), as the CFPS data show that a notable share of male divorces occurs between the ages of 40 and 50.

Tables 9 and 10 present results using the census and the CFPS data, respectively. The sex ratio coefficients are positive and significant in all columns except for 2005, when divorce was still uncommon. These findings indicate that higher sex ratios raise men's divorce rates.

We also examine the effect of the sex ratios on men's remarriage rates, with results reported in Appendix Table B.5. The coefficients are negative, suggesting that the higher sex ratios reduce the likelihood of remarriage for men, although the estimates are not statistically significant. The negative signs are consistent with the outside option channel: after divorce, men reenter a male-skewed marriage market, where increased competition makes remarriage more difficult.

Faced with a higher risk of divorce, how do men respond to preserve their marriages? The existing research and our findings suggest that men and their families invest more to improve women's marital satisfaction, such as purchasing better housing. Women also

Table 9: Gender Imbalance and Men's Probability of Divorce: Censuses

Divorce	Pooled (1)	Census 2005 (2)	Census 2010 (3)	Census 2015 (4)
Sex ratio	0.019***	0.003	0.022***	0.027***
	(0.004)	(0.005)	(0.005)	(0.006)
Obs.	1,661,579	449,980	858,222	353,377
Adj. R-sq	0.012	0.012	0.011	0.012
Mean of Dep. Var.	0.025	0.019	0.026	0.033

NOTE: All columns control for prefecture-by-wave and birth year-by-wave fixed effects, as well as fixed effects for education and number of children. Standard errors clustered at the prefecture level are shown in parentheses. ***p<0.01, *p<0.05, p<0.1.

Table 10: Gender Imbalance and Men's Probability of Divorce: CFPS 2010

1{Ever divorced}	(1)	(2)	(3)	(4)
Sex ratio	0.077*	0.072**	0.075**	0.076**
	(0.041)	(0.035)	(0.035)	(0.035)
Obs.	8,517	8,517	8,456	8,455
Adj. R-sq	0.018	0.075	0.076	0.076
Mean of Dep. Var.	0.043	0.043	0.043	0.043

NOTE: All columns control for fixed effects for prefecture, birth year, education, and number of children. Standard errors clustered at the prefecture level are shown in parentheses. ***p<0.01, *p<0.05, p<0.1.

report higher marital satisfaction, likely reflecting greater efforts by men. We provide direct evidence that men adjust their domestic behavior in response to the higher sex ratios. As shown in Table 11, men spend more time caring for family members on weekdays. However, results for other outcomes are inconclusive. We find no significant effects on men's housework, employment, or working hours.

Table 11: Gender Imbalance and Men's Time Spent Caring for Family Members

Time spent caring for family members	Hours on weekdays	IHS(Hours) on weekdays	Hours on weekends	IHS(Hours) on weekends
	(1)	(2)	(3)	(4)
Sex ratio	0.536**	0.318**	0.206	0.200
	(0.217)	(0.132)	(0.344)	(0.167)
Obs.	8,511	8,511	8,503	8,503
Adj. R-sq	0.066	0.094	0.074	0.108
Mean of Dep. Var.	0.596	0.421	0.786	0.523

NOTE: This table shows estimations using the CFPS data. Only men who are currently married are included in the sample. Prefecture, birth year, education, and number of children fixed effects are controlled for all columns. All the control variables for the census regressions and the indicator for cohabitation before marriage are controlled for in all columns. Standard errors clustered at the prefecture level are reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table 12: Gender Imbalance and Husband's Mental Health

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
The frequency of feeling	Upset	Stressed	Cannot calm down	Hopeless	Everything is difficult	Life is meaningless	Overall depression
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Sex ratio	0.298	0.337**	0.223	-0.015	0.534**	0.037	0.699*
	(0.200)	(0.169)	(0.187)	(0.154)	(0.221)	(0.140)	(0.388)
Obs.	8,234	8,236	8,236	8,230	8,235	8,236	8,220
Adj. R-sq	0.053	0.032	0.044	0.031	0.053	0.041	0.065
Mean of Dep. Var.	0.635	0.491	0.417	0.287	0.538	0.253	-0.221

NOTE: This table reports results using CFPS data, restricted to currently married men. Outcomes in Columns (1)–(6) are indices measuring the frequency of depressive symptoms, ranging from 0 (none of the time) to 4 (all of the time). Column (7) reports the first principal component from a Principal Component Analysis (PCA) of these measures. All columns control for fixed effects for prefecture, birth year, education, and number of children, as well as the full set of controls used in the census regressions and an indicator for pre-marital cohabitation. Standard errors clustered at the prefecture level are shown in parentheses. ***p<0.01, *p<0.05, p<0.1.

If men are making efforts to preserve their marriages, why do divorce rates still rise? According to the Coase Theorem, if a utility surplus exists and transaction costs are low, men should be able to transfer enough surplus to prevent divorce. We provide suggestive evidence that this surplus may be shrinking. Table 12 shows that the higher sex ratios are associated with poorer mental health among men, suggesting they experience greater psychological stress in male-skewed markets.

Overall, the results suggest that men's marital surplus is significantly diminished. Higher male-to-female ratios are associated with higher divorce rates. Evidence on men's mental health suggests that men experience greater psychological distress in male-skewed markets, likely reflecting reduced bargaining power at home.

6 Conclusion

Although the birth control policy has officially ended, its long-lasting effects will continue to shape the Chinese economy. This paper highlights a previously overlooked pattern: the strong link between sex imbalance and divorce. Our analysis shows that the outside option channel plays a central role. Women in male-skewed markets are more likely to remarry after divorce and to remarry higher-quality partners than those in more balanced markets. The expectation of easier remarriage and better prospects makes women more willing to leave unsatisfying marriages.

Given that the outside option channel is the main mechanism, regions with imbalanced sex ratios may face similar marriage instability. Our framework provides a useful lens for understanding gender imbalances that arise from incarceration, migration, or war. This study underscores the potential long-term impacts of such shocks on the marriage prospects of entire cohorts.

We acknowledge that other factors may shape how sex ratios influence divorce. While sex ratios can affect both marriage quality and the availability of alternative options, the dominant force likely depends on context. Social norms, legal institutions, and social support all play a role. Such variation may explain the mixed findings in the literature on gender imbalance and divorce. This paper does not aim to establish a universal rule for when the outside option channel dominates. Instead, it offers complementary evidence from a unique setting of female scarcity and aims to inspire further research on gender-based differences in marriage preferences and behavior.

A A Model on Marriage, Divorce, and Re-marriage

In this section, we present a simple model to highlight how the sex ratio may impact the divorce rate. The impact is ambiguous. On the one hand, a higher sex ratio can increase the marriage quality, which decreases divorce rates; on the other hand, a higher sex ratio can increase outside options, which destabilizes marriage. The model demonstrates the impacts of the two forces, illustrating that it is possible to derive either positive or negative outcomes from the same theoretical framework under varying conditions. The model is related to previous papers such as Burdett and Mortensen (1998) where workers search both on and off the job.

We assume that all women are identical, and we consider the decisions of an infinitely lived woman in continuous time. At a moment in time, each woman is either single (state 0) or married (state 1). At random time intervals, a woman meets a new potential partner. The arrival rates depend on a woman's current state, i.e., single or married. Let λ_i, $i \in \{0, 1\}$ represent the parameter of a Poisson arrival process, which denotes the arrival rate of new partners while a woman is currently in state i.

A higher sex ratio means higher arrival rates of new potential partners for both single and married women. Namely, λ_0 and λ_1 increase with the sex ratio. In the following analyses, we first derive comparative statics with respect to λ_0 and λ_1. Then, we discuss the implications with different relative impacts of the sex ratio on λ_0 and λ_1.

Women are assumed to search among potential partners randomly. The quality of a new partner is assumed to be the realization of a random draw from a distribution F if a woman is single and a distribution G if a woman is married. Women must respond to new partners as soon as they arrive; there is no recall.

Given the framework briefly outlined above, the expected discounted lifetime utility when a woman is single, U, can be expressed as the solution to the following equation:

$$rU = b + \lambda_0 \int \max\{V(w) - U, 0\} dF(w)$$

In other words, the opportunity cost of searching while single, the interest on its asset value, is equal to utility while single plus the expected capital gain attributable to searching for an acceptable partner where acceptance occurs only if the value of marriage, $V(w)$, exceeds that of continued search.

Similarly, the expected lifetime utility of a married woman currently with a partner of quality w solves the following:

$$rV(w) = w + \lambda_1 \int \max\{X(\tilde{w} - c) - V(w), 0\} dG(\tilde{w})$$

where $X(\tilde{w}-c)=\frac{\tilde{w}-c}{r}$. We assume that people divorce at most once, potentially due to the substantial divorce cost.[4]

We solve the model backward. For married women with a husband of quality w, there exists a reservation quality, w_{R1}, such that $X(w_{R1}-c)=V(w)$. w_{R1} satisfies the following:

$$w_{R1}=c+w+\frac{\lambda_1}{r}\int_{w_{R1}}(\tilde{w}-w_{R1})dG(\tilde{w}) \tag{5}$$

Equation 5 is the implicit function expression of w_{R1}. w_{R1} is thus a function of w and λ_1. c and r also affect w_{R1}, and we assume that they are exogenous and take them as given. Denote w_{R1} as $w_{R1}(w,\lambda_1)$. It can be derived that:

$$\frac{\partial w_{R1}(w,\lambda_1)}{\partial w}=\frac{1}{1+\frac{\lambda_1}{r}(1-G(w_{R1}))}>0$$

$$\frac{\partial w_{R1}(w,\lambda_1)}{\partial \lambda_1}=\frac{\frac{1}{r}\int_{w_{R1}}(\tilde{w}-w_{R1})dG(\tilde{w})}{1+\frac{\lambda_1}{r}(1-G(w_{R1}))}>0$$

For single women, there exists a reservation quality, w_{R0}, such that $V(w_{R0})=U$. w_{R0} satisfies the following:

$$rU=rV(w_{R0})=w_{R1}(w_{R0},\lambda_1)-c=b+\frac{\lambda_0}{r}\int_{w_{R0}}(w_{R1}(w,\lambda_1)-w_{R1}(w_{R0},\lambda_1))dF(w)$$

Rearrange the equation, we get the implicit function that defines the value of w_{R0}:

$$w_{R1}(w_{R0},\lambda_1)=c+b+\frac{\lambda_0}{r}\int_{w_{R0}}(w_{R1}(w,\lambda_1)-w_{R1}(w_{R0},\lambda_1))dF(w) \tag{6}$$

It can be derived that:

$$\frac{\partial w_{R0}}{\partial \lambda_0}=\frac{\frac{1}{r}\int_{w_{R0}}(w_{R1}(w,\lambda_1)-w_{R1}(w_{R0},\lambda_1))dF(w)}{\frac{\partial w_{R1}(w_{R0},\lambda_1)}{\partial w_{R0}}\left(1+\frac{\lambda_0}{r}(1-F(w_{R0}))\right)}>0$$

which means a higher arrival rate in the first-marriage market increases the reservation value.

[4]This assumption is consistent with data. According to CFPS 2010 data, less than 1% of people ever married more than twice.

Additionally, it can be derived that:

$$\frac{\partial w_{R0}}{\partial \lambda_1} = -\frac{\frac{\partial w_{R1}(w_{R0},\lambda_1)}{\partial \lambda_1} - \frac{\lambda_0}{r}\int_{w_{R0}}\left(\frac{\partial w_{R1}(w,\lambda_1)}{\partial \lambda_1} - \frac{\partial w_{R1}(w_{R0},\lambda_1)}{\partial \lambda_1}\right) dF(w)}{\frac{\partial w_{R1}(w_{R0},\lambda_1)}{\partial w_{R0}}\left(1+\frac{\lambda_0}{r}(1-F(w_{R0}))\right)} < 0 \tag{7}$$

We then analyze the duration of the first marriage, which is related to divorce. Specifically, a longer duration of the first marriage corresponds to a lower divorce rate. The average duration of the first marriage has the following expression:

$$E(\text{duration}) = \int_{w_{R0}} \frac{1}{\lambda_1(1-G(w_{R1}(w,\lambda_1)))} dF(w)/[1-F(w_{R0})]$$

It can be derived that:

$$\begin{aligned}\frac{\partial E(\text{duration})}{\partial \lambda_0} =& \frac{\partial w_{R0}}{\partial \lambda_0}\frac{f(w_{R0})}{(1-F(w_{R0}))^2}\frac{1}{\lambda_1}\times \\ & \int_{w_{R0}}\left(\frac{1}{1-G(w_{R1}(w,\lambda_1))} - \frac{1}{1-G(w_{R1}(w_{R0},\lambda_1))}\right) dF(w)\end{aligned} \tag{8}$$

It can be shown that $\frac{\partial E(\text{duration})}{\partial \lambda_0} > 0$.

$$\begin{aligned}\frac{\partial E(\text{duration})}{\partial \lambda_1} =& \int_{w_{R0}} \frac{\partial \frac{1}{\lambda_1(1-G(w_{R1}(w,\lambda_1)))}}{\partial \lambda_1} dF(w)\frac{1}{1-F(w_{R0})} \\ &+ \frac{\partial w_{R0}}{\partial \lambda_1}\frac{f(w_{R0})}{(1-F(w_{R0}))^2}\frac{1}{\lambda_1}\times \\ & \int_{w_{R0}}\left(\frac{1}{1-G(w_{R1}(w,\lambda_1))} - \frac{1}{1-G(w_{R1}(w_{R0},\lambda_1))}\right) dF(w)\end{aligned} \tag{9}$$

It can be shown that $\frac{\partial E(\text{duration})}{\partial \lambda_1} < 0$ if the distribution $G(\tilde{w})$ is log-concave.

The equation suggests that there are two forces through which λ_1 affects the duration of the first marriage. Firstly, λ_1 is the arrival rate and affects the duration of the first marriage directly. Secondly, expecting a high arrival rate after marriage, women lower their reservation value for the first marriage, which affects the duration. In other words, λ_1 affects the duration indirectly by affecting the reservation value for the first marriage, w_{R0}.

Denote the sex ratio as s, $\eta_0 = \frac{\partial \lambda_0}{\partial s}$ and $\eta_1 = \frac{\partial \lambda_1}{\partial s}$, we can derive the following:

$$\frac{\partial w_{R0}}{\partial s} = \frac{\partial w_{R0}}{\partial \lambda_0}\eta_0 + \frac{\partial w_{R0}}{\partial \lambda_1}\eta_1$$

Therefore, we have $\frac{\partial w_{R0}}{\partial s} > 0$ as long as $\frac{\eta_0}{\eta_1} > -\frac{\frac{\partial w_{R0}}{\partial \lambda_1}}{\frac{\partial w_{R0}}{\partial \lambda_0}}$. Given that most people are in their first marriages and the observed average quality of marriages is the average of qualities above w_{R0}, namely $E[w|w > w_{R0}]$, $\frac{\partial w_{R0}}{\partial s} > 0$ means the overall marriage quality increases with a higher sex ratio.

Additionally, Equation (8) suggests that $\frac{\partial E(\text{duration})}{\partial \lambda_0}$ can be expressed as $A\frac{\partial w_{R0}}{\partial \lambda_0}$ where A is a positive number. Equation (9) suggests that $\frac{\partial E(\text{duration})}{\partial \lambda_1}$ can be expressed as $(A\frac{\partial w_{R0}}{\partial \lambda_1} - B)$ where B is also a positive number. Combined together, we can derive the following:

$$\frac{\partial E(\text{duration})}{\partial s} = \frac{\partial E(\text{duration})}{\partial \lambda_0}\eta_0 + \frac{\partial E(\text{duration})}{\partial \lambda_1}\eta_1 = A\left(\frac{\partial w_{R0}}{\partial \lambda_0}\eta_0 + \frac{\partial w_{R0}}{\partial \lambda_1}\eta_1\right) - B\eta_1$$

Therefore, we have $\frac{\partial E(\text{duration})}{\partial s} < 0$ as long as $\frac{\eta_0}{\eta_1} < \frac{B/A - \frac{\partial w_{R0}}{\partial \lambda_1}}{\frac{\partial w_{R0}}{\partial \lambda_0}}$. $\frac{\partial E(\text{duration})}{\partial s} < 0$ means the duration of first marriages decreases when the sex ratio increases, and thus divorce rates increase with a higher sex ratio.

Given that $\frac{\partial w_{R0}}{\partial \lambda_1} < 0$, there exist values of η_0 and η_1 that satisfy both $\frac{\eta_0}{\eta_1} > -\frac{\frac{\partial w_{R0}}{\partial \lambda_1}}{\frac{\partial w_{R0}}{\partial \lambda_0}}$ and $\frac{\eta_0}{\eta_1} < \frac{B/A - \frac{\partial w_{R0}}{\partial \lambda_1}}{\frac{\partial w_{R0}}{\partial \lambda_0}}$.

The framework presented in this section implies that, with certain parameters, we can observe both: 1) the marriage quality increases with the sex ratio, and 2) the divorce rates increase with the sex ratio.

B Additional Tables and Figures

B.1 Descriptive Statistics

Table B.1: Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max
Panel A: Province-level Data				
(# of divorce cases / population aged 15-64)×1000‰	2.503	2.118	.001	9.479
(# of divorce cases / # of marriage cases)×100	28.449	14.96	6.872	80.812
Sex ratio	1.056	.051	.946	1.224
ln(GDP per capita)	9.947	.937	7.719	12.009
Average household size	3.258	.485	2.222	6.79
Average year of education	8.309	1.375	2.948	12.782
Panel B: Census Data				
Divorce	.016	.127	0	1
Sex ratio	1.031	.077	.826	1.259
Han Ethnicity	.913	.282	0	1
Employed	.775	.417	0	1
Migrant	.269	.444	0	1
Urban	.353	.478	0	1
Having a son	.651	.477	0	1
Number of children	1.297	.762	0	9
Panel C: Individual Survey Data				
Divorce	.028	.166	0	1
Sex ratio	1.039	.065	.842	1.325
Han Ethnicity	.912	.283	0	1
Employed	.753	.431	0	1
Migrant	.093	.291	0	1
Urban	.472	.499	0	1
Cohabit	.122	.327	0	1
Having a son	.692	.462	0	1
Number of children	1.581	.851	0.000	7
Number of male sibling(s)	1.485	1.181	0.000	7
Number of female sibling(s)	1.364	1.302	0.000	8
Income	8348.388	17044.929	0.000	800000

NOTE: The province-level data has 744 observations except for the divorce-marriage ratio, which has 651 observations. The census data has 1,150,344 observations; the CFPS survey data has 8,717 observations.

B.2 Robustness Tests

Table B.2: Robustness Tests

	Age 25-45	[-6,+1]	[-8,+2]/[-7,+3]	Wave-specific sex ratio
	(1)	(2)	(3)	(4)
Sex Ratio	0.022***	0.016***	0.020***	0.021***
	(0.006)	(0.004)	(0.004)	(0.005)
Obs.	1,387,342	1,150,344	1,150,344	1,231,916
Adj. R-sq	0.020	0.018	0.018	0.018
Mean of Dep. Var.	0.019	0.016	0.016	0.016

NOTE: This table shows the results of the first set of robustness tests. Column (1) reports the estimations using the sample of women aged 25-45. Columns (2) and (3) report results using [-6, +1] and [-8,+2]/[-7,+3] age gaps for the calculation of the sex ratio. Column (4) shows the results using census wave-specific sex ratio. Specifically, we construct the sex ratio by aggregating data from both the present and preceding waves of censuses. For example, to construct the sex ratio for observations in the 2010 census, we calculate the sex ratio pooling 2000, 2005, and 2010 waves. Standard errors clustered at the prefecture level are reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table B.3: Robustness Tests (Cont.)

	Industry & occupation	Province × year FE	Fake divorce	Exclude migrants	County-level sex ratio
	(1)	(2)	(3)	(4)	(5)
Sex Ratio	0.018***	0.016***	0.019***	0.017***	0.013***
	(0.005)	(0.004)	(0.004)	(0.005)	(0.003)
Obs.	891,729	1,150,346	1,150,346	840,431	632,769
Adj. R-sq	0.021	0.018	0.018	0.020	0.018
Mean of Dep. Var.	0.016	0.016	0.016	0.017	0.019

NOTE: This table shows the results of the second set of robustness tests. Column (1) reports the estimations controlling for census wave-specific occupation and industry fixed effects. Column (2) reports the results controlling for province × year fixed effects. Column (3) reports results controlling for the implementation of fake divorce. Column (4) shows the results excluding migrants. Column (5) shows the results using an alternative sex ratio measure. Specifically, we restrict the sample to the census years 2010 and 2015 because the year 2005 does not have information on the residential county. For 2010 and 2015, we replaced the prefecture-level sex ratio with the county-level sex ratio if the women lived in rural areas. Standard errors clustered at the prefecture level are reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

B.3 Results using Instrumental Variables

We have experimented with three types of instruments. First, previous literature has documented that the land reform in China in the 1980s has increased the male-to-female ratio (Almond et al., 2019). We use the indicator for whether the land reform has begun in the women's living prefecture five years before her birth year as the instrument for the sex ratio in her birth cohort.

Secondly, the spread of ultrasound machines facilitated sex selection (see, e.g. Hesketh and Xing, 2006). We use the indicator for the presence of ultrasound machines in the capital of each woman's province of residence in her birth year as an instrument for the sex ratio.[5]

Finally, as migrants remain primarily men in our sample period, increases in migration can decrease the origin's sex ratio. Correspondingly, we use the foreign direct investment (FDI) in the main flow-in prefectures in China: Beijing, Shanghai, and Shenzhen, weighted by the inverse distance to the women's prefecture of residence as an instrument for sex ratio. Specifically, for women living in prefecture p born in year y, we construct the following instrument:

$$IV_{py} = \sum_{c} \frac{lnFDI_{c,y+20}}{distance_{pc}}$$

where c can be Beijing, Shanghai, or Shenzhen, $lnFDI_{c,y+20}$ is the log of FDI in prefecture c in year $y+20$, $distance_{pc}$ is the distance in miles between prefecture p and c. The sample excludes women living in Beijing, Shanghai, and Shenzhen. The first stage is expected to be negative as migration decreases the origin's sex ratio.

Table B.4 presents the results using all three instruments. The IVs have relatively weak first stages. Only lnFDI/distance has an F-stat larger than 10, the rule of thumb. While the second-stage estimations are noisy, all the confidence intervals cover the OLS estimation. We have experimented with various variations for all these instruments, including age thresholds, distance measures, etc. The results remain quantitatively similar.

[5]The data on the presence of ultrasound machines comes from Almond et al. (2019).

Table B.4: IV Results

	Sex ratio (1)	Divorce (2)	Sex ratio (3)	Divorce (4)	Sex ratio (5)	Divorce (6)
Sex ratio		0.119** (0.055)		0.012 (0.046)		0.044** (0.020)
Land Reform	0.027*** (0.008)					
Ultrasound			0.014** (0.006)			
lnFDI/distance					-5.499*** (1.339)	
F-stat	11.04		5.467		16.865	
Obs.	1,071,983	1,150,344	1,048,586	1,048,586	892,060	892,060

NOTE: Odd columns are the first stages, and even columns are the second stages for IV estimations. Standard errors clustered at the prefecture level are reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

B.4 Gender Imbalance and Remarriage of Men

Table B.5: Gender Imbalance and Remarriage of Men

1{Ever-remarry}	Divorced	Divorced & Widowed
	(1)	(2)
Sex ratio	-1.186	-0.909
	(0.799)	(0.737)
Obs.	366	483
Adj. R-sq	0.224	0.194
Mean of Dep. Var.	0.488	0.447

NOTE: Prefecture, birth year, education, number of children, male and female siblings fixed effects are controlled for all columns. All the control variables for the census regressions and the indicator for cohabitation before marriage are controlled for all columns. Standard errors clustered at the prefecture level are reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

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